

Primitive Arrays

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Topics list

- Why arrays?
- Primitive Arrays
- Array Syntax
- Loops and Arrays
- What datatypes can be stored in arrays?
- Difference between fixed size and number of populated data

Why arrays?

- We look at different pieces of code to explain the concept.
- In each piece of code, we:
 - read in 10 numbers from the keyboard
 - add the numbers
 - print the sum of all the numbers.

Adding 10 numbers

```
import java.util.Scanner;
:
Scanner input = new Scanner(System.in);
:
int n;
int sum = 0;

for (int i = 0; i < 10; i++) {
    n = input.nextInt();
    sum += n;
}
```

Reads in 10 numbers
from the keyboard

```
System.out.println("The sum of the values you typed in is : " +  
sum);
```

Adding 10 numbers

```
import java.util.Scanner;
:
Scanner input = new Scanner(System.in);
:
int n;
int sum = 0;

for (int i = 0; i<10; i++) {
    n = input.nextInt();
    sum += n;
}
```

As each number is entered,
it is added to the value
currently stored in **sum**.

```
System.out.println("The sum of the values you typed in is : " + sum);
```

Adding 10 numbers

```
import java.util.Scanner;
:
Scanner input = new Scanner(System.in);
:
int n;
int sum = 0;

for (int i = 0; i<10; i++) {
    n = input.nextInt();
    sum += n;
}
```

When the 10 numbers have
been read in,
the **sum** of the 10 numbers is
printed to the console.

```
System.out.println("The sum of the values you typed in is : " + sum);
```

Adding 10 numbers

```
import java.util.Scanner;
:
Scanner input = new Scanner(System.in);
:
int n;
int sum = 0;

for (int i = 0; i < 10; i++) {
    n = input.nextInt();
    sum += n;
}
```

Notice that,
each time a number is read in,
it overwrites the value stored in **n**.

It doesn't remember
the individual numbers typed in.

```
System.out.println("The sum of the values you typed in is : " + sum);
```

Rule – Never lose input data



- Always try to **store** input data for later use
- In real-life systems,
you nearly always need to use it again.
- The previous code has NOT done this.
 - Let's try another way ...

Remembering the Numbers

```
int n0,n1, n2, n3, n4, n5, n6, n7, n8, n9;  
int sum = 0;  
  
n0 = input.nextInt();  
sum += n0;  
  
n1 = input.nextInt();  
//rest of code for n2 to n8  
  
n9= input.nextInt();  
sum += n9;  
  
System.out.println("The sum of the val
```

This works in the sense that we have retained the input data.

BUT...we no longer use loops.

Imagine the code if we had to read in 1,000 numbers?

We need a new approach...

This is where **data structures** come in!

We will now look at **arrays**.

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- What datatypes can be stored in arrays?
- Difference between fixed size and number of populated data

Arrays (fixed-size collections)

- Arrays are a way to collect associated values.
- Programming languages usually offer a special **fixed-size collection** type: an *array*.
- Java arrays can store
 - objects
 - primitive-type values.
- Arrays use a special syntax.

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Primitive types

Primitive type

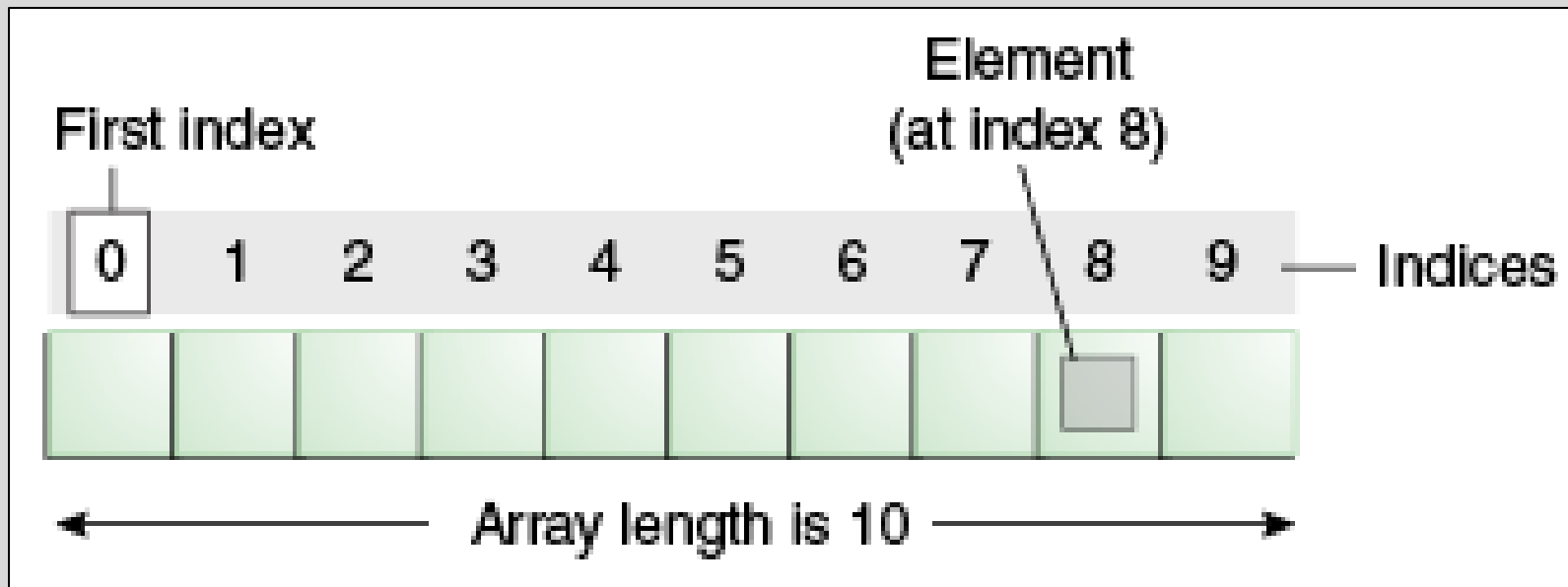
```
int num = 17;
```

Directly stored
in memory...

17

- We are now going to look at a **structure** that can **store many values** of the **same type**.
- Imagine a structure made up of sub-divisions or sections...
- Such a structure is called an **array** and would look like:

Structure of a primitive array



Structure of a primitive array

int[] numbers;

numbers

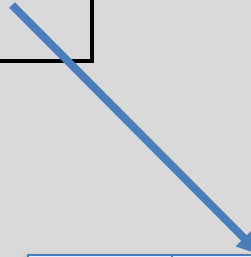
null

Structure of a primitive array

```
int[] numbers;
```

```
numbers = new int[4];
```

numbers



0	0
1	0
2	0
3	0

Structure of a primitive array

```
int[] numbers;
```

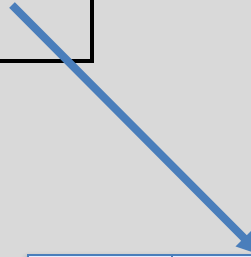
```
numbers = new int[4];
```

We have declared an array of `int`, with a capacity of four.

Each element is of type **int**.

The array is called **numbers**.

numbers



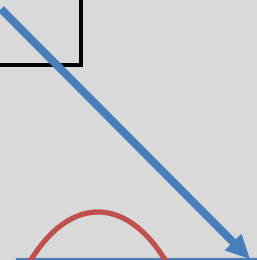
0	0
1	0
2	0
3	0

Structure of a primitive array

```
int[] numbers;
```

```
numbers = new int[4];
```

numbers



0	0
1	0
2	0
3	0

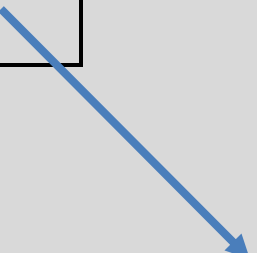
Index of each
element in the array

Structure of a primitive array

```
int[] numbers;
```

```
numbers = new int[4];
```

numbers



0	0
1	0
2	0
3	0

Default value for each
element of type **int**.

Structure of a primitive array

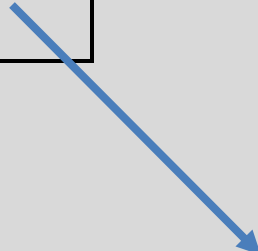
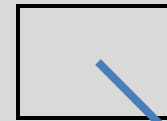
```
int[] numbers;
```

```
numbers = new int[4];
```

```
numbers[2] = 18;
```

We are directly
accessing the
element at index **2**
and setting it to a
value of **18**.

numbers



0	0
1	0
2	18
3	0

Structure of a primitive array

```
int[] numbers;
```

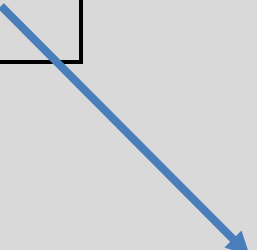
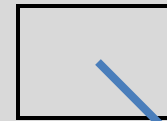
```
numbers = new int[4];
```

```
numbers[2] = 18;
```

```
numbers[0] = 12;
```

We are setting the
element at index **0**
and to a value of **12**.

numbers



0	12
1	0
2	18
3	0

Structure of a primitive array

```
int[] numbers;
```

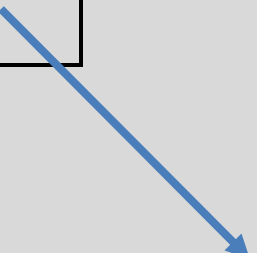
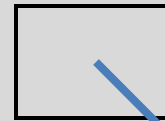
```
numbers = new int[4];
```

```
numbers[2] = 18;
```

```
numbers[0] = 12;
```

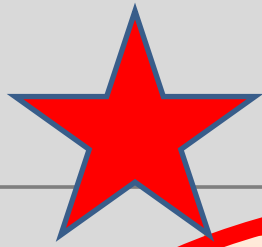
```
System.out.println(numbers[2]);
```

numbers



0	12
1	0
2	18
3	0

Here we are printing the contents of
index location 2
i.e. 18 will be printed to the console.

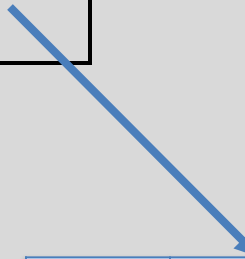
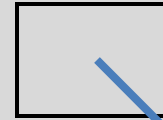


Declaring a primitive array

```
int[] numbers;  
//somecode  
numbers = new int[4];
```

This is how we
previously
declared our
array of four **int**,
called **numbers**.

numbers



0	0
1	0
2	0
3	0

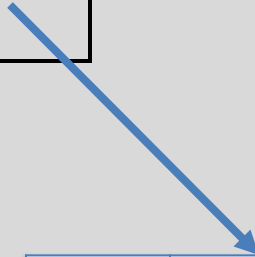
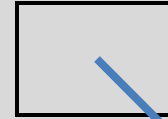
Declaring a primitive array

```
int[] numbers;  
//somecode  
numbers = new int[4];
```

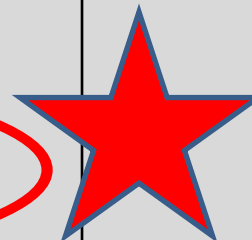
We can also
declare it like
this...

```
int[] numbers = new int[4];
```

numbers



0	0
1	0
2	0
3	0



Returning to our method
that reads in, and sums, 10 numbers
(typed in from the keyboard)...

and converting it to use primitive arrays...

Version that doesn't save the numbers

```
import java.util.Scanner;
:
Scanner input = new Scanner(System.in);
:
int n;
int sum = 0;
```

```
for (int i = 0; i < 10; i++) {
    n = input.nextInt();
    sum += n;
}
```

```
System.out.println("The sum of the values you typed in is : " + sum);
```

Notice that,
each time a number is read in,
it overwrites the value stored in **n**.

It doesn't remember
the individual numbers typed in.

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- Why arrays?
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Using arrays to remember numbers

```
import java.util.Scanner;
:
Scanner input = new Scanner(System.in);
:
int numbers[] = new int[10];
int sum = 0;
```

```
//read in the data
for (int i = 0; i < 10 ; i ++ ) {
    numbers[i] = input.readInt();
}
```

```
// now we sum the values
for (int i = 0; i < 10 ; i ++ ) {
    sum += numbers[i];
}
```

```
System.out.println("The sum of the values you typed in is : " + sum);
```

Using an array
to store each value
that was entered.

Using arrays to remember numbers

```
import java.util.Scanner;
:
Scanner input = new Scanner(System.in);
:
int numbers[] = new int[10];
int sum = 0; Loop 1
```

```
//read in the data
for (int i = 0; i < 10 ; i ++ ) {
    numbers[i] = input.readInt();
}
```

Loop 2

```
// now we sum the values
for (int i = 0; i < 10 ; i ++ ) {
    sum += numbers[i];
}
```

```
System.out.println("The sum of the values you typed in is : " + sum);
```

Q: Can we reduce the code to only have **one loop**?

Could we move the “sum” code into the first loop?

Using arrays to remember numbers

```
import java.util.Scanner;
:
Scanner input = new Scanner(System.in);
:
int numbers[] = new int[10];
int sum = 0;
```

```
//read in the data
for (int i = 0; i < 10 ; i ++) {
    numbers[i] = input.readInt();
    sum += numbers[i];
}
```

```
System.out.println("The sum of the values you typed in is : " + sum);
```

A: Yes.

Move the “sum” code into the first loop.

-> functionality doesn't change

Loop 1

What if we wanted the user
to decide how many numbers
they wanted to sum?

```
import java.util.Scanner;
:
Scanner input = new Scanner(System.in);
int sum = 0;
```

```
//Using the numData value to set the size of the array
int numbers[];
System.out.println("How many numbers do you need?");
```

```
int numData = input.nextInt();
```

```
numbers = new int [numData];
```

```
//read in the data and sum the values
for (int i = 0; i < numData ; i ++ ) {
    numbers[i] = input.nextInt();
    sum += numbers[i];
}
```

```
System.out.println("The sum of the values you typed in is : " + sum);
```

1. Delclare **numbers** to be an array of type integer.
2. **numData** takes in the size.
3. Use numData to initialize the array with **new** specifying the size.

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What type of data
can be stored
in a primitive array?

An array can store ANY TYPE of data.

Primitive Types

```
int numbers[] = new int[10];
```

```
byte smallNumbers[] = new byte[4];
```

```
char characters[] = new char[26];
```

Object Types

```
String words = new String[30];
```

```
Person persons[] = new Person[20];
```

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Difference between fixed size and number of populated data

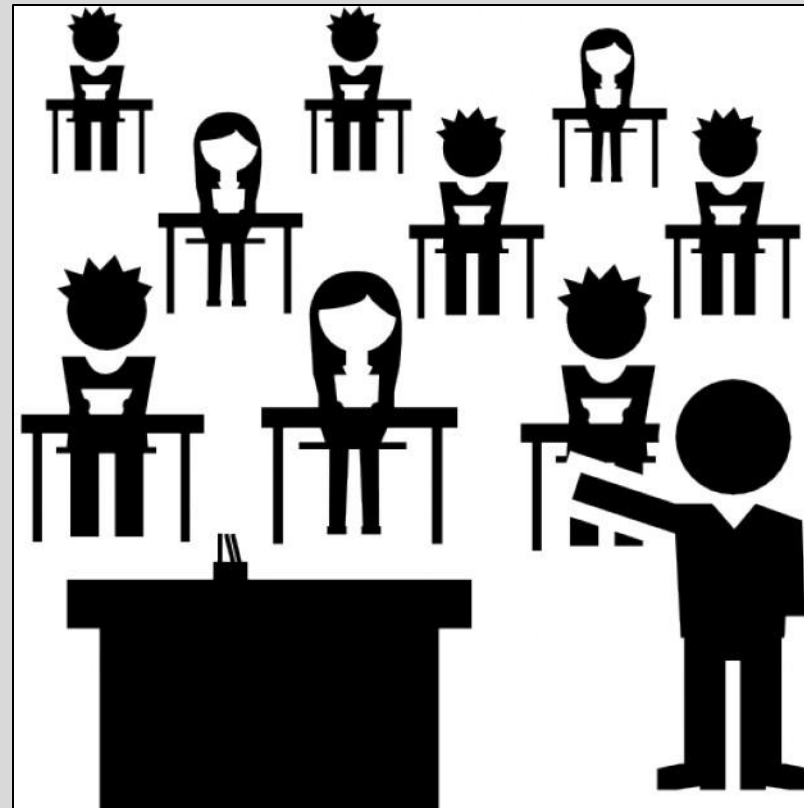
Do we have to use **all** the elements in the array?

Do we have to use all elements in the array?

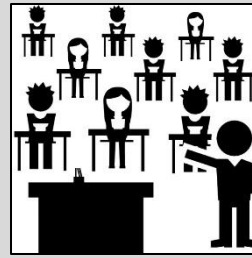
- No.
- **But**...this might cause logic errors, if we don't take this into consideration in our coding.
- Consider this scenario...

Scenario – exam results and **average grade**

- We have a class of 15 students.
- They have a test coming up.
- We want to store the results in an array and then find the average result.



Average grade



 **results**

0	56
1	65
2	45
3	78
4	98
5	41
6	40
7	55
8	45
9	51
10	42
11	78
12	0
13	0
14	0

We create an array of int with a capacity of 15

Only 12 students sat the exam. Their results were recorded in the first 12 elements

To calculate the average result, divide by the number of **populated elements** **NOT** the array capacity.

Do we have to use all elements in the array?

- If all elements in an array are NOT populated, we need to:
 - have another variable (e.g. int **size**)
 - containing the number of elements in the array **actually used**.
 - ensure size is used when processing the array
 - e.g.
`for (int i= 0; i < size; i++)`
- For now though, we assume that all elements of the array are populated and therefore ready to be processed.

Summary - Arrays

- Arrays are structures that can store many values of the same type
- Rule – Never lose input data
 - Arrays enable us to store the data efficiently
 - We can use loops with arrays
- Arrays can store ANY type
- Declaring arrays

```
int[] arryName;  
//somecode  
arryName= new int[4];
```

OR

```
int[] arryName= new int[4];
```

Questions?

